Dirac-Kronig-Penney Model
Holes
Effective mass

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Dirac-Kronig-Penney model

It is discussed in the previous section that the energy band structure calculations require numerous solutions of the time-independent Schrödinger equations (5.1.41) for various wave vectors k within the first Brillouin zone. This is usually done numerically, and special techniques have been developed for this purpose. Most notable among them are the pseudo-potential method and the density functional method. It is apparent that band structure calculations are laborious and computer-intensive. It this section, we illustrate the energy band calculations by considering the simplest version of one-dimensional Kronig-Penney model in which the periodic potential $U(x)$ is modeled by periodically spaced Dirac delta functions (see Figure 5.3). For such a potential, the analytical treatment is possible, and the

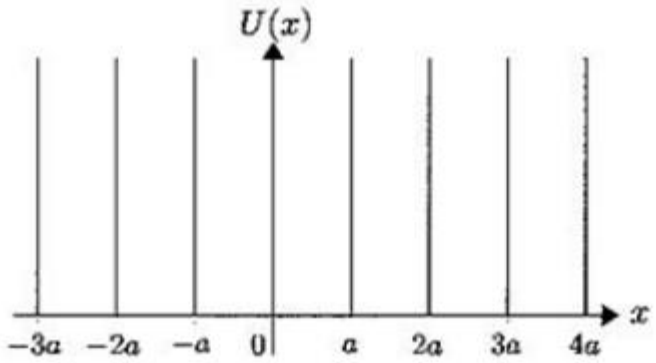


Fig. 5.3

main features of energy band structures discussed in the previous section can be vividly illustrated.

In our analysis, we shall use the following Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x), \tag{5.2.1}$$

where

$$U(x) = \beta \sum_{\nu=-\infty}^{\infty} \delta(x - \nu a), \tag{5.2.2}$$

and β is some constant.

It is apparent that

$$U(x + a) = U(x). \tag{5.2.3}$$

The time-independent Schrödinger equation for the above potential can be written as follows

$$-\frac{\hbar^2}{2m}\frac{d^2\Phi(x)}{dx^2} + \beta\left(\sum_{\nu=-\infty}^{\infty} \delta(x - \nu a)\right)\Phi(x) = \mathcal{E}\Phi(x). \tag{5.2.4}$$

According to formula (5.1.26), any solution of this equation has the following property

$$\Phi(x + a) = e^{ika}\Phi(x), \tag{5.2.5}$$

which implies that if $\Phi(x)$ is known on the interval

$$0 < x < a, \tag{5.2.6}$$

then by using the last formula it can be found for any x, i.e., it can be extended to the entire x-axis. For this reason, it is sufficient to find $\Phi(x)$ on the above interval. It is clear that in the above interval equation (5.2.4) can be written as follows

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\Phi(x)}{dx^2} = \mathcal{E}\Phi(x),} \quad (0 < x < a). \tag{5.2.7}$$

We shall next find the boundary conditions for $\Phi(x)$ at $x = 0_+$ and $x = a_-$, where subscripts "+" and "−" imply that the boundary points of the interval (5.2.6) are approached from "above" and "below," respectively, i.e., from within the interval.

Since $\Phi(x)$ is a continuous function, we have

$$\Phi(a_+) = \Phi(a_-). \tag{5.2.8}$$

According to formula (5.2.5), we have

$$\Phi(a_+) = e^{ika}\Phi(0_+). \tag{5.2.9}$$

By substituting the last formula into equation (5.2.8), we end up with the following boundary condition

$$\boxed{e^{ika}\Phi(0_+) = \Phi(a_-).} \tag{5.2.10}$$

These boundary conditions relate the values of $\Phi(x)$ at the ends of the interval (5.2.6).

Next, we shall derive the boundary condition that relates the values of $\frac{d\Phi}{dx}(x)$ at the ends of the same interval. The derivation proceeds as follows.

Consider the interval

$$0 < x < 2a. \tag{5.2.11}$$

On this interval, equation (5.2.4) has the form

$$-\frac{\hbar^2}{2m}\frac{d^2\Phi(x)}{dx^2} + \beta\delta(x-a)\Phi(x) = \mathcal{E}\Phi(x). \tag{5.2.12}$$

By integrating both sides of the last equation over the very small interval $a-\alpha < x < a+\alpha$, we obtain

$$-\frac{\hbar^2}{2m}\left[\frac{d\Phi}{dx}(a+\alpha) - \frac{d\Phi}{dx}(a-\alpha)\right] + \beta\Phi(a_+) = \mathcal{E}\int_{a-\alpha}^{a+\alpha}\Phi(x)dx. \tag{5.2.13}$$

In the limit when $\alpha \to 0$, from the last equation we find

$$\frac{d\Phi}{dx}(a_+) - \frac{d\Phi}{dx}(a_-) = \frac{2m\beta}{\hbar^2}\Phi(a_+). \tag{5.2.14}$$

From formula (5.2.5), it follows that

$$\frac{d\Phi}{dx}(a_+) = e^{ika}\frac{d\Phi}{dx}(0_+). \tag{5.2.15}$$

By substituting formulas (5.2.9) and (5.2.15) into equation (5.2.14), we end up with the boundary condition

$$\boxed{e^{ika}\frac{d\Phi}{dx}(0_+) - \frac{d\Phi}{dx}(a_-) = \frac{2m\beta}{\hbar^2}e^{ika}\Phi(0_+).} \tag{5.2.16}$$

Thus, in order to find $\Phi(x)$ in the interval (5.2.6) we have to find a solution of equation (5.2.7) subject to the boundary conditions (5.2.10) and (5.2.16).

A general solution of equation (5.2.7) can be written in the form

$$\Phi(x) = A\cos qx + B\sin qx, \quad (0 < x < a), \tag{5.2.17}$$

where

$$q = \frac{\sqrt{2m\mathcal{E}}}{\hbar}. \tag{5.2.18}$$

It is easy to derive the following equations for A and B from the boundary conditions (5.2.10) and (5.2.16), respectively,

$$e^{ika}A = A\cos qa + B\sin qa, \tag{5.2.19}$$

$$qa[e^{ika}B + A\sin qa - B\cos qa] = 2\gamma e^{ika}A, \tag{5.2.20}$$

where

$$\gamma = \frac{m\beta a}{\hbar^2}. \tag{5.2.21}$$

From equation (5.2.19), we find

$$\frac{A}{B} = \frac{\sin qa}{e^{ika} - \cos qa}. \tag{5.2.22}$$

Similarly, from equation (5.2.20) it follows that

$$\frac{A}{B} = \frac{e^{ika} - \cos qa}{\frac{2\gamma}{qa}e^{ika} - \sin qa}. \tag{5.2.23}$$

From the last two equations, we conclude that

$$\frac{\sin qa}{e^{ika} - \cos qa} = \frac{e^{ika} - \cos qa}{\frac{2\gamma}{qa}e^{ika} - \sin qa}, \tag{5.2.24}$$

which yields

$$(e^{ika} - \cos qa)^2 - \sin qa\left(\frac{2\gamma}{qa}e^{ika} - \sin qa\right) = 0. \tag{5.2.25}$$

Further algebraic transformations lead to

$$e^{i2ka} - 2e^{ika}\left(\cos qa + \gamma\frac{\sin qa}{qa}\right) + 1 = 0. \tag{5.2.26}$$

Dividing the last equation by $2e^{ika}$, we find

$$\cos qa + \gamma\frac{\sin qa}{qa} = \frac{e^{ika} + e^{-ika}}{2}, \tag{5.2.27}$$

which leads to the final equation

$$\boxed{\cos qa + \gamma\frac{\sin qa}{qa} = \cos ka.} \tag{5.2.28}$$

For any given k, the last equation can be solved numerically to find the corresponding values of q and, according to formula (5.2.18), the corresponding

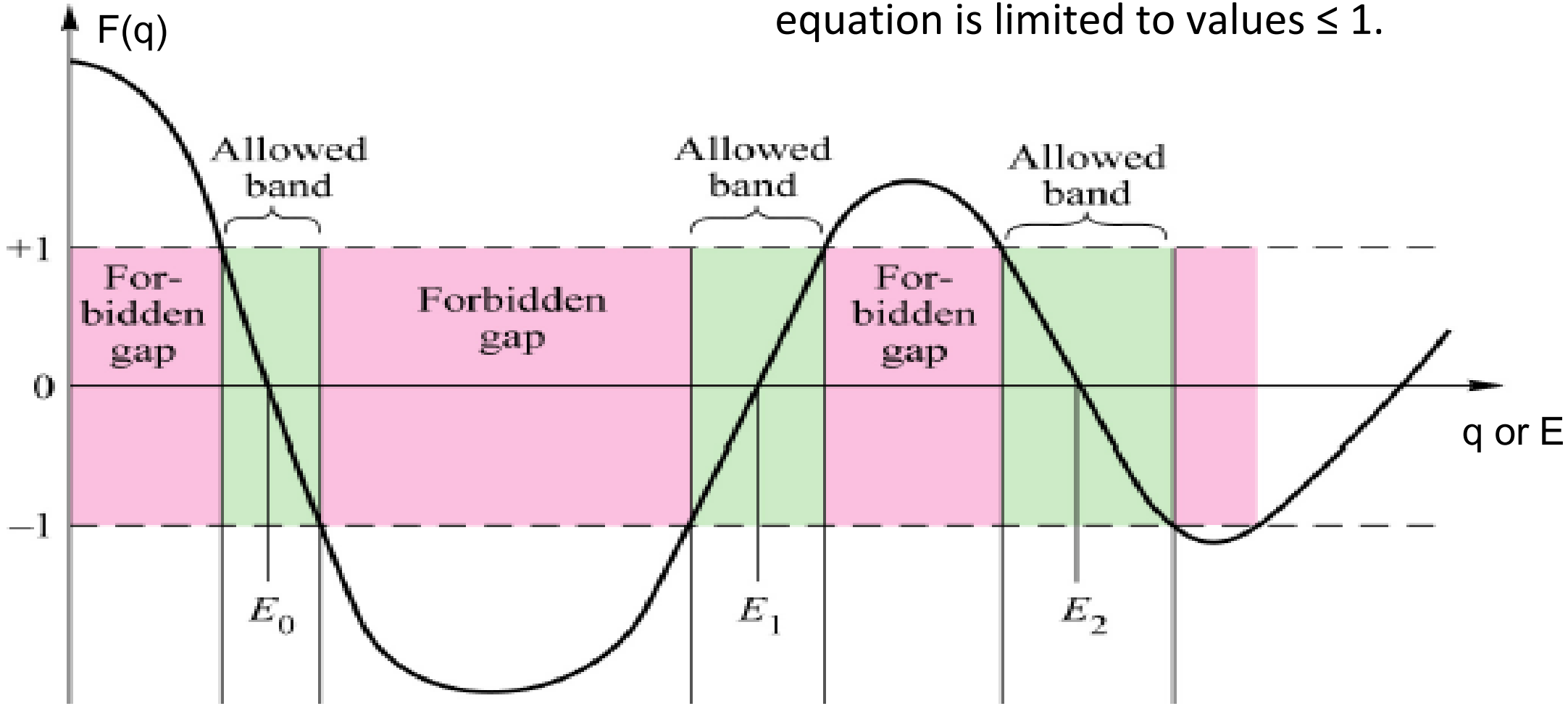
values of $\mathcal{E}$. In this way, the band structures $\mathcal{E}_n(k)$ can be computed. The described solution process can be illustrated graphically and the emergence of energy bands becomes clearly visible. To this end, we introduce the function

$$F(q) = \cos qa + \gamma \frac{\sin qa}{qa} \tag{5.2.29}$$

and write the equation (5.2.28) in the form

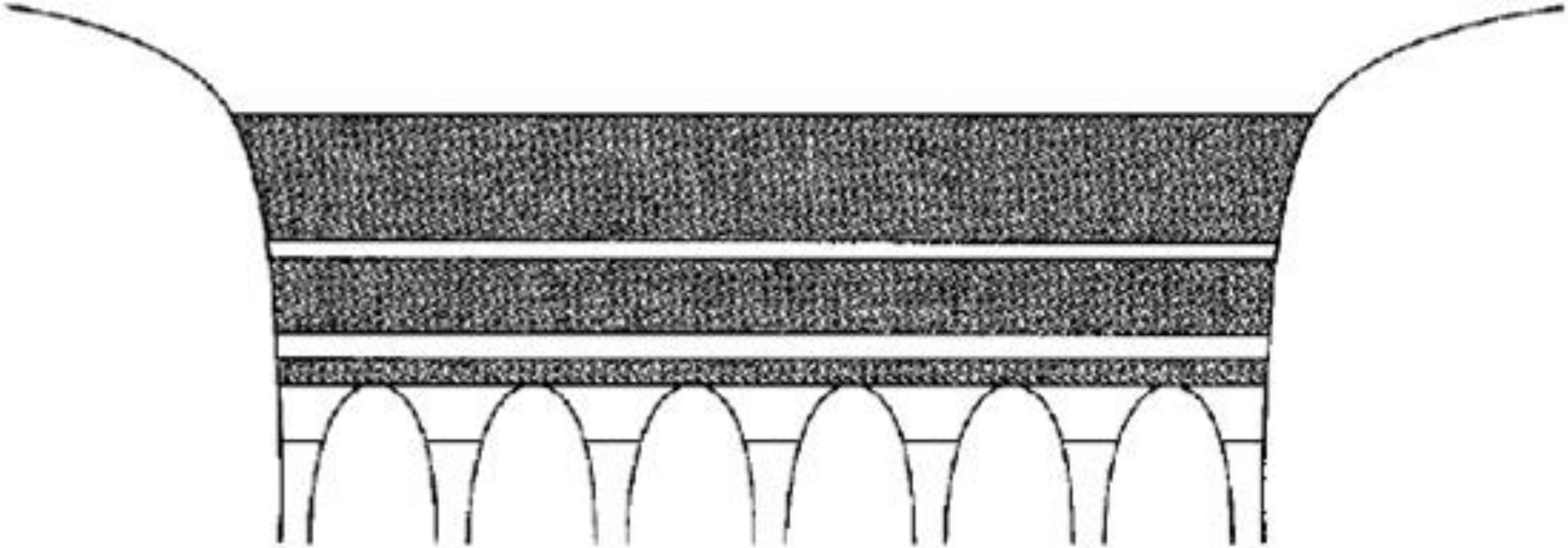
$$F(q) = \cos ka. \tag{5.2.30}$$

F(q) has a solution, only if $|F(q)| \leq 1$ because the cosine function on the right-hand side of the equation is limited to values ≤ 1.



F.E Schubert, Physical foundations of solid state devices

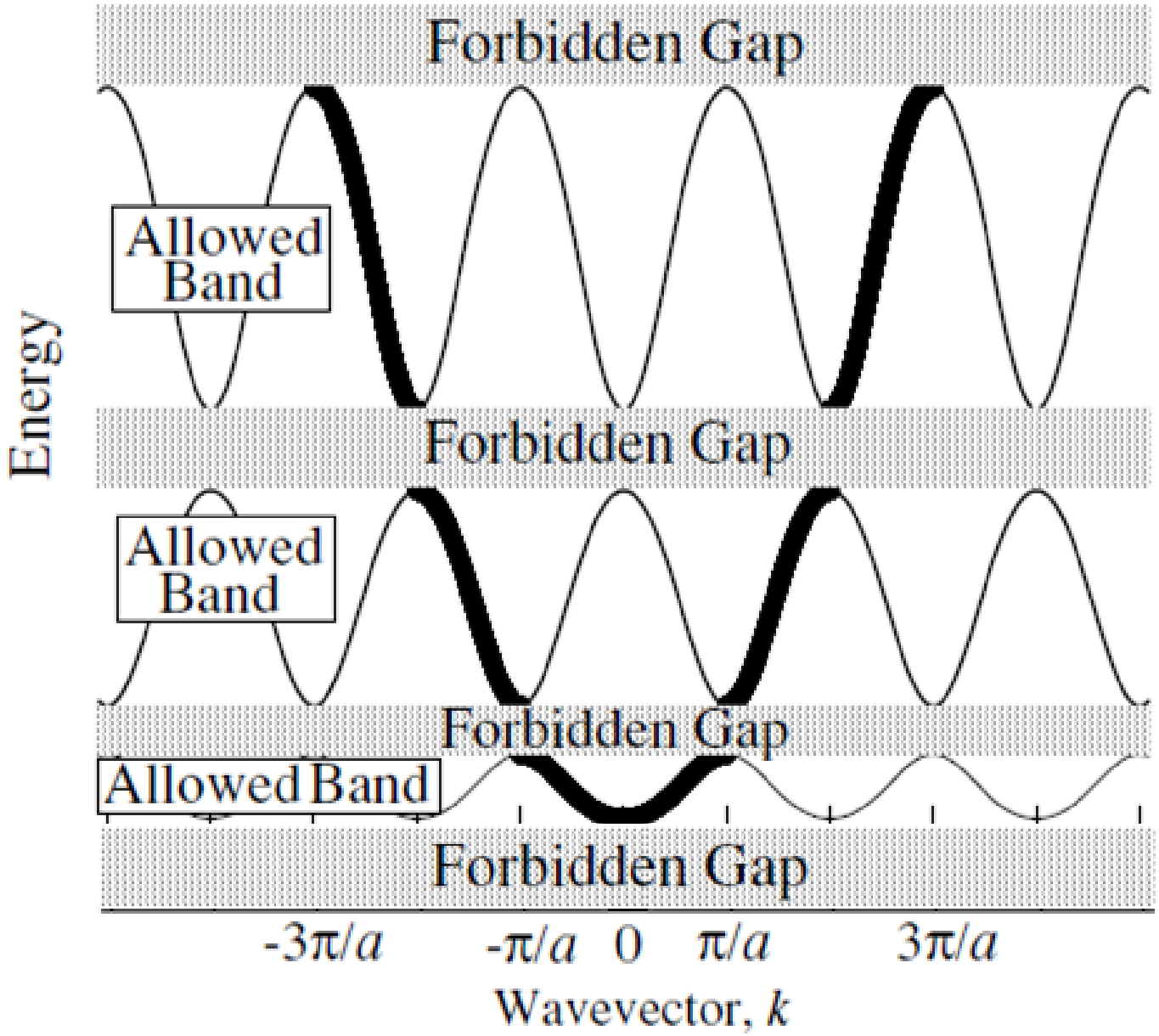
Emergence of energy bands and gaps



No discrete energy levels as in isolated atoms, but consists of allowed bands and forbidden gaps.

Energy bands naturally arise from the solution of Schrodinger equation in a periodic potential well

Periodic E-k diagram



Please note that these aren't sinusoidal functions in general. It is periodic with a maxima and a minima

Laue's condition

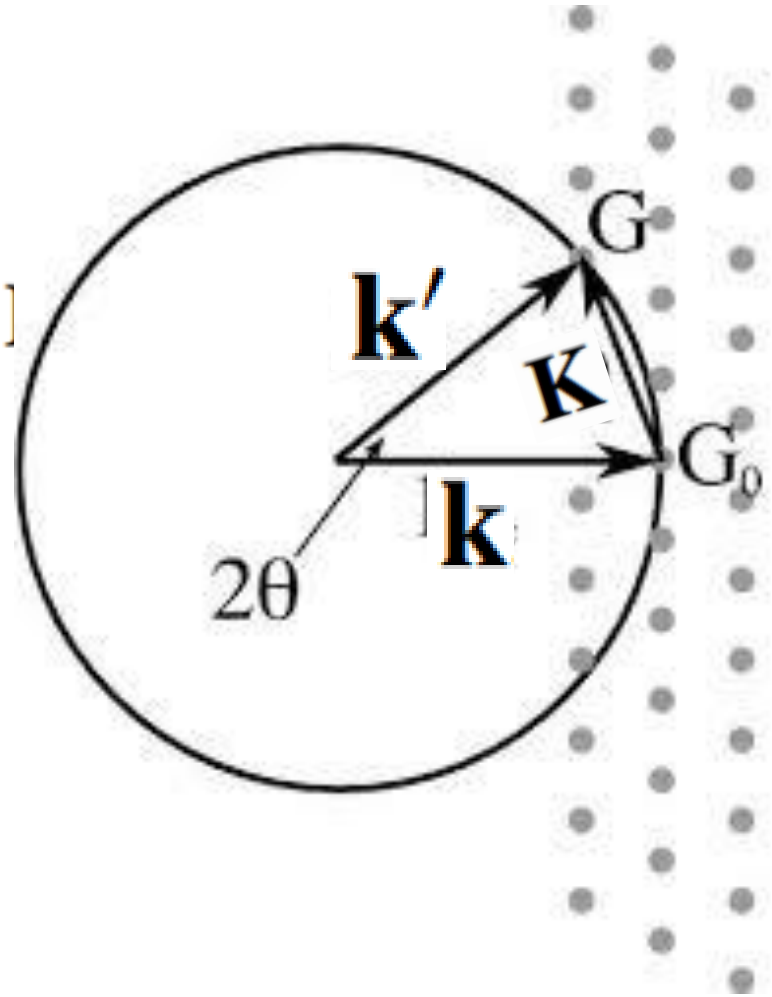
$$\mathbf{R} \cdot (\mathbf{k} - \mathbf{k}') = 2\pi m$$

$$e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}} = 1, \quad \text{for all Bravais lattice vectors } \mathbf{R}$$

Comparing the condition with definition of reciprocal lattice vector,

$$\mathbf{K} = \mathbf{k}' - \mathbf{k}$$

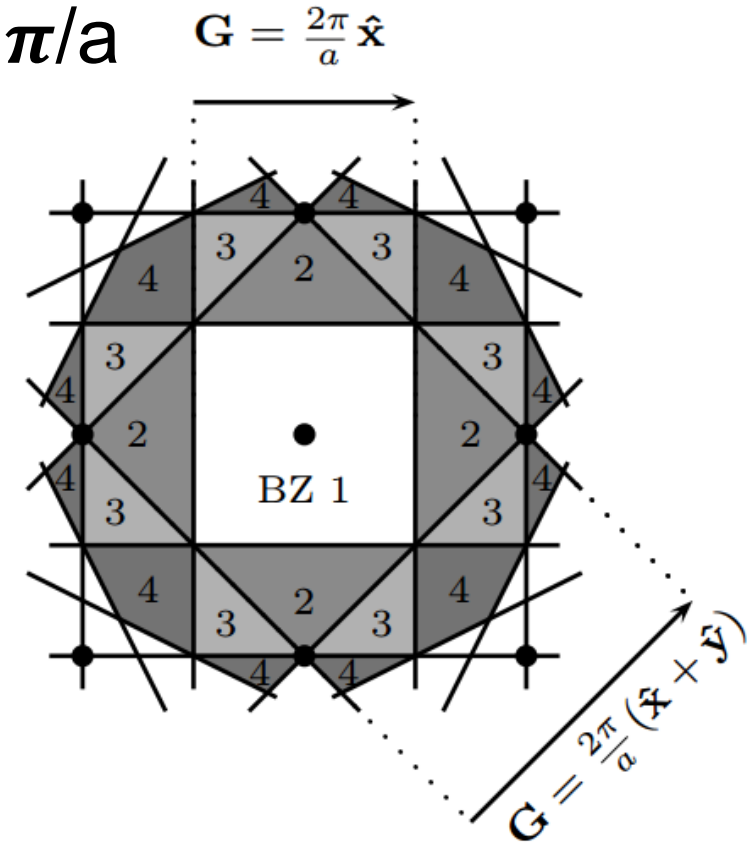
Constructive interference will occur provided that the change in wave vector is a vector of reciprocal lattice



Ewald sphere in reciprocal space

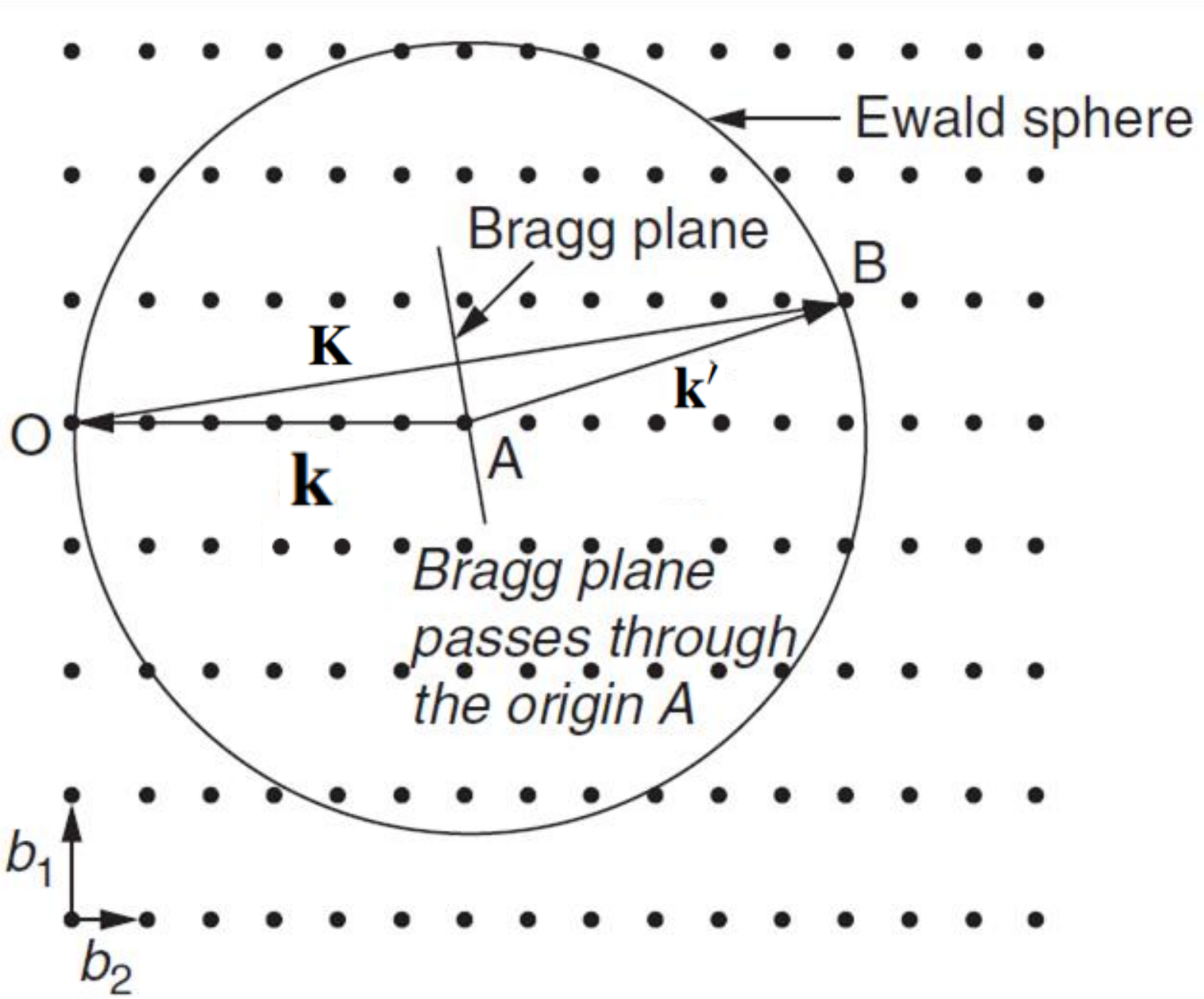
Brillouin zone

- 1st **Brillouin zone** is the Wigner-Seitz primitive cell of the reciprocal lattice
- Since waves are physically equivalent under shifts of the wavevector k by a reciprocal lattice vector $2\pi/a$ (G), we can do all representation within the first Brillouin zone (BZ)
- N^{th} Brillouin zone (BZ) is the region in reciprocal space between n^{th} and $(n-1)^{th}$ **Bragg planes** (perpendicular bisectors to the line joining the origin and any reciprocal lattice points)

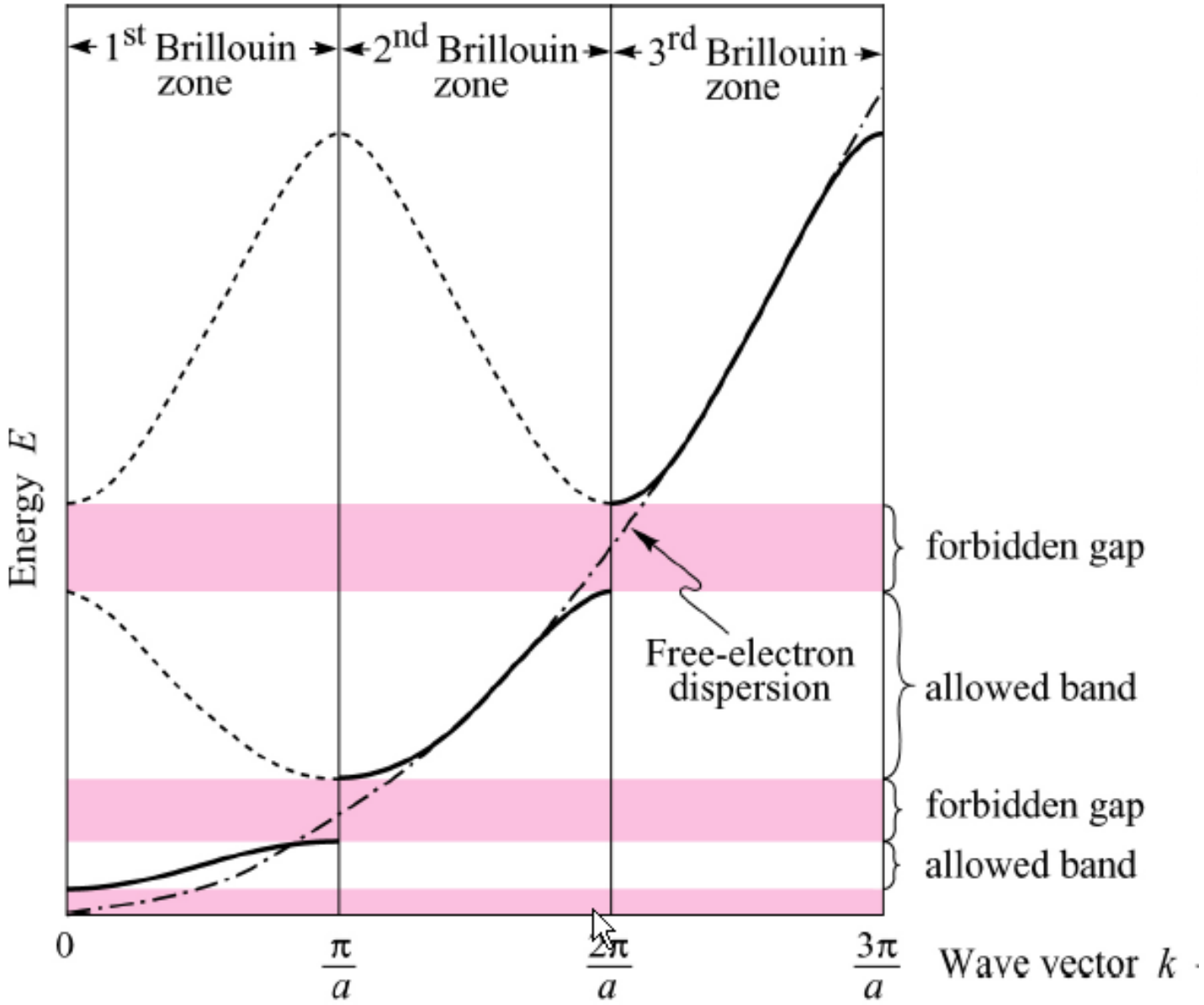


Diffraction in Brillouin zone boundary

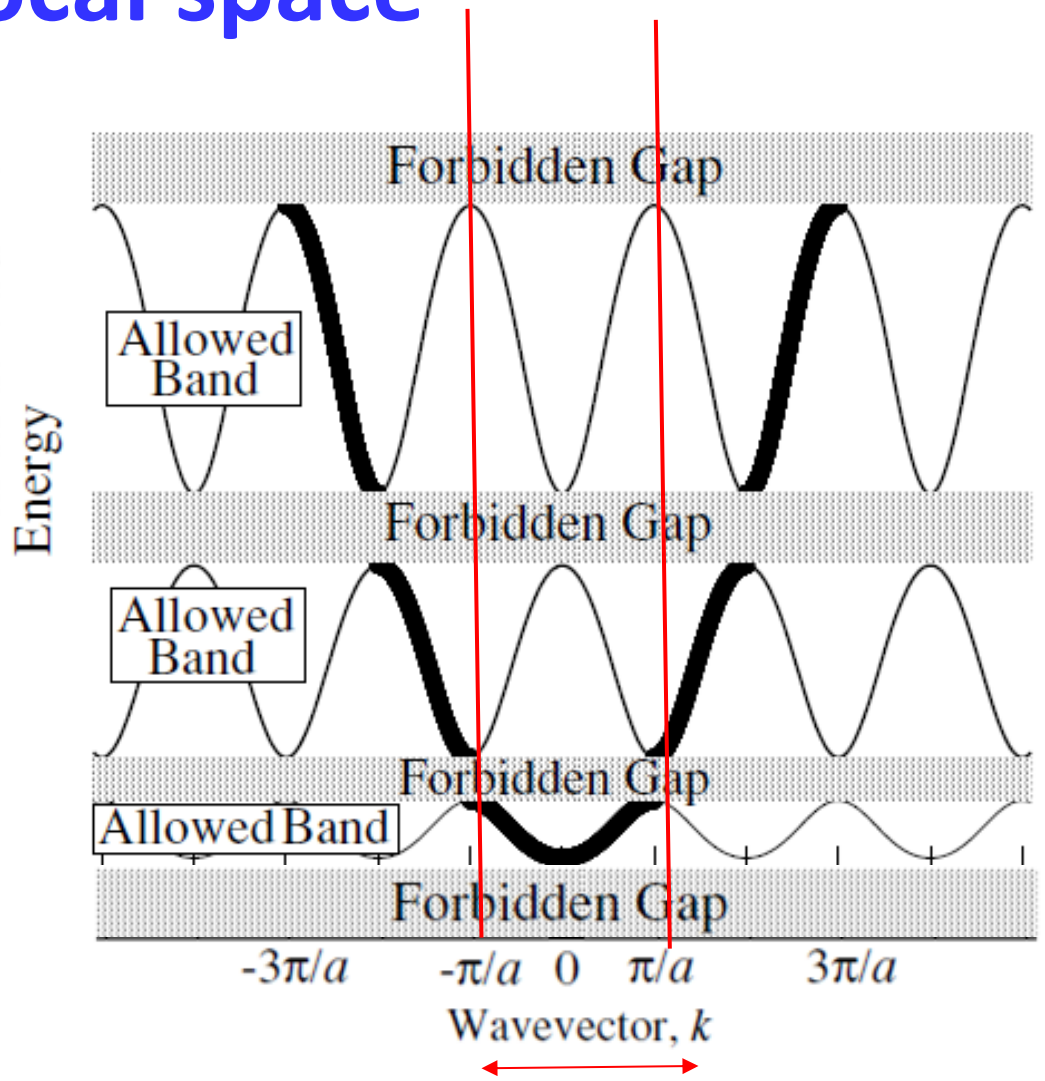
- Whenever a Bragg plane touches one end of a wave vector, the condition for diffraction is satisfied
- Diffraction occurs whenever a wave vector touches BZ boundary



Energy bands and gaps in reciprocal space

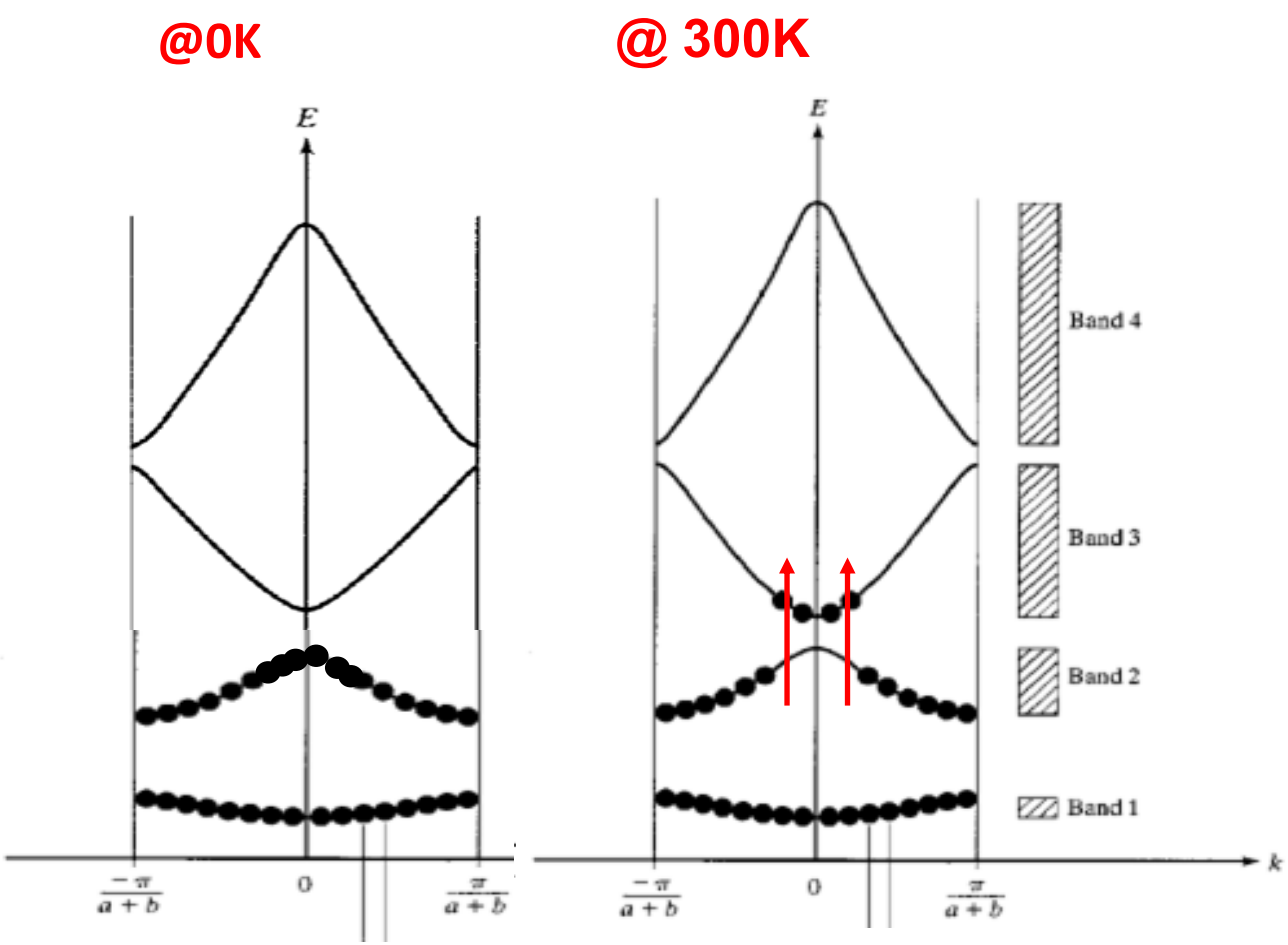


Extended Zone



Reduced Zone

How bands are occupied by electrons?



Top most completely filled band @ 0K – valence band (VB)

Band above VB – conduction band (CB)

At higher temp, electrons can gain energy and move to higher bands, leaving behind vacancies.

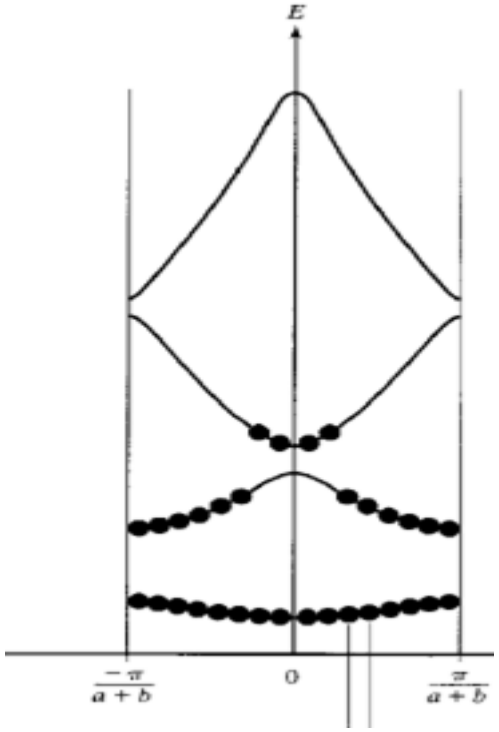
Completely filled bands don't contribute to current due to symmetry of E-k

$$v(k) = \frac{1}{\hbar}\frac{dE}{dk}$$

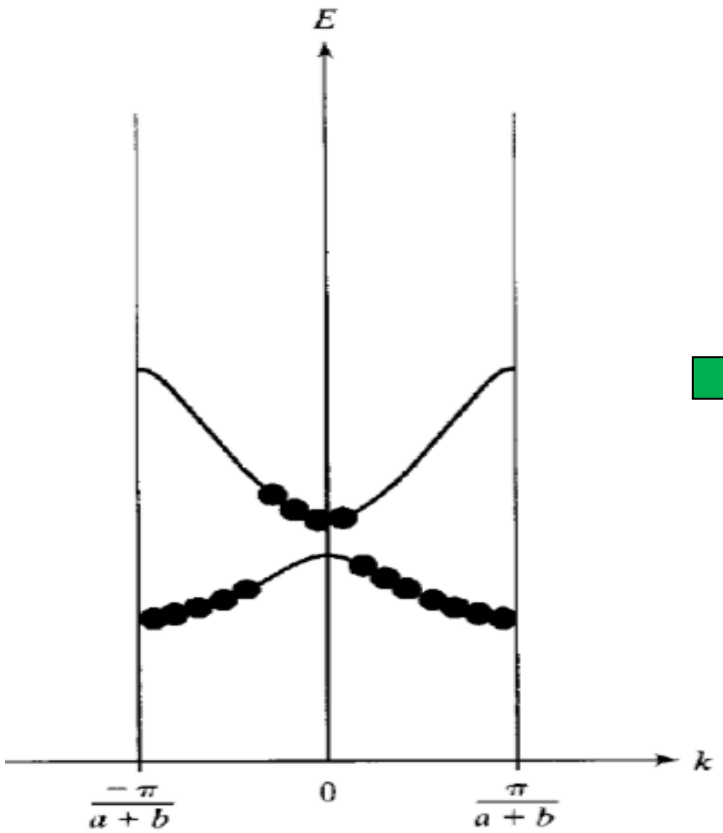
$$v(-k) = -\,v(k)$$

Partially occupied band & Concept of hole

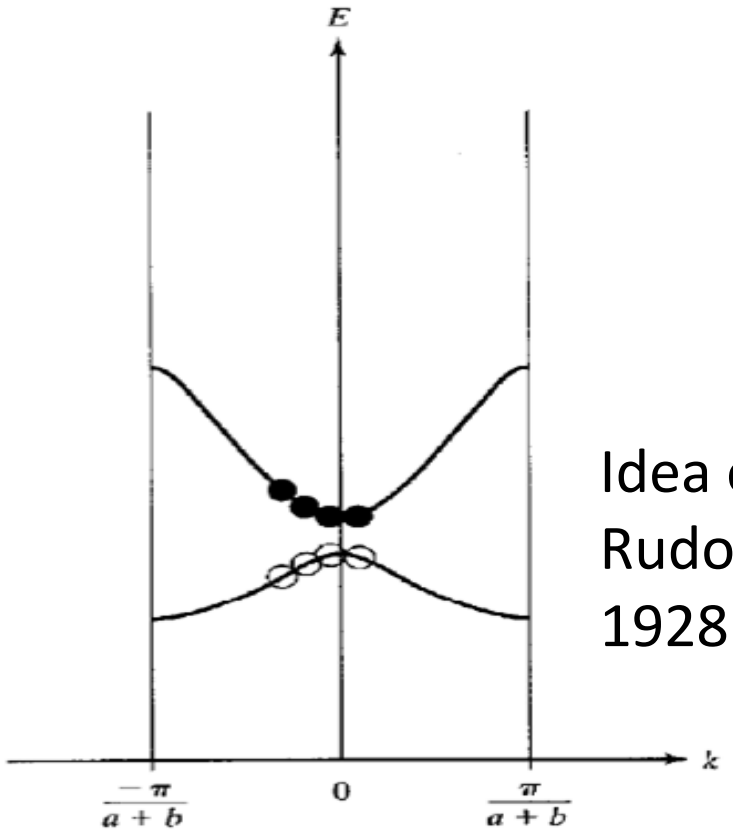
Zero field @300K



Electric field in +ve x direction



(a)

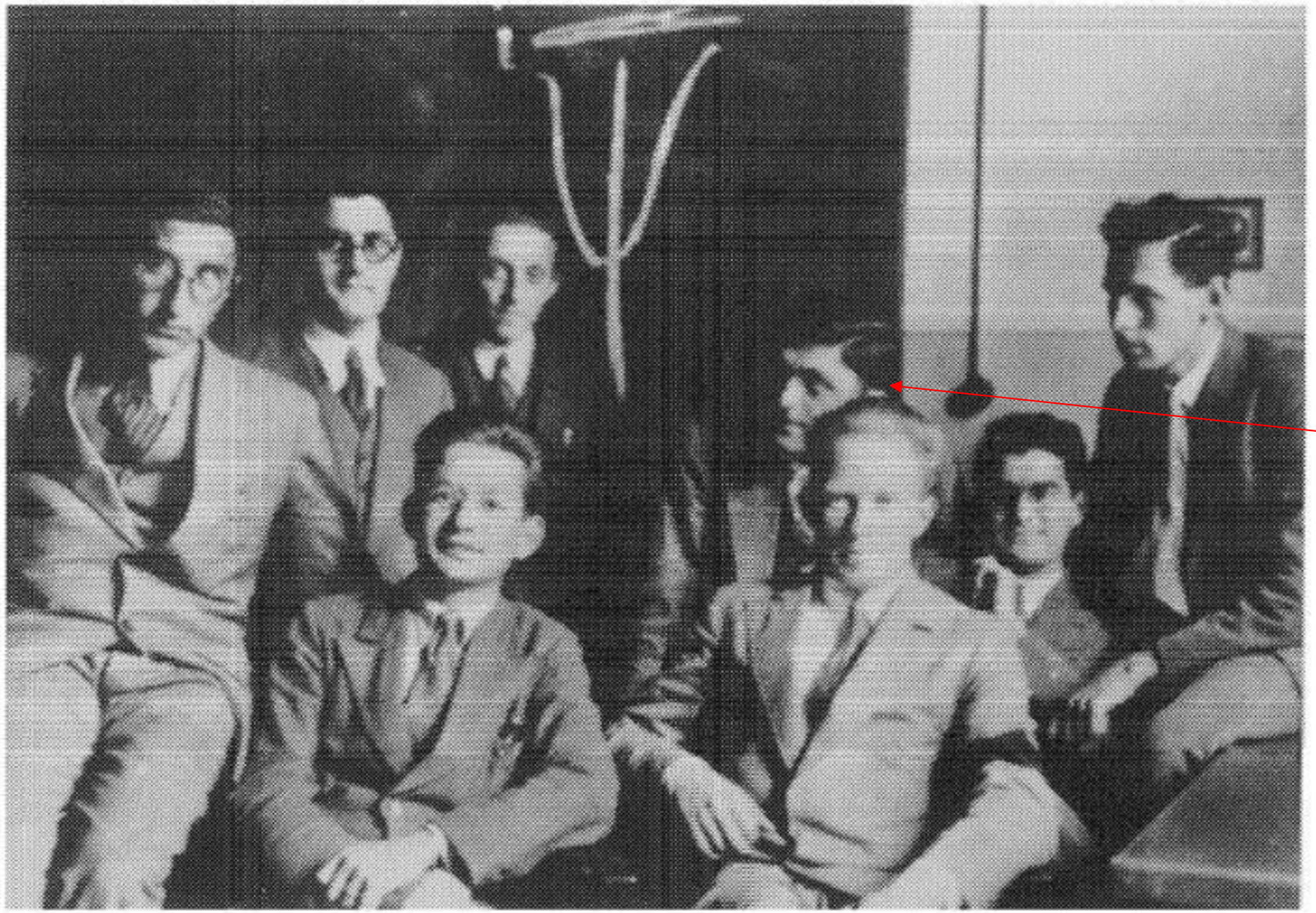


(b)

Idea of holes - Rudolf Peierls in 1928

$$J_{vb} = \frac{1}{V}\left(\sum_{\text{all states}}(-q)v_i - \sum_{\text{empty states}}(-q)v_i\right) \qquad J_{vb} = \frac{1}{V}\sum_{\text{empty states}}(+q)v_i$$

Dynamics of large number of VB electrons with negative charge ≡ small number of vacant states with positive charge



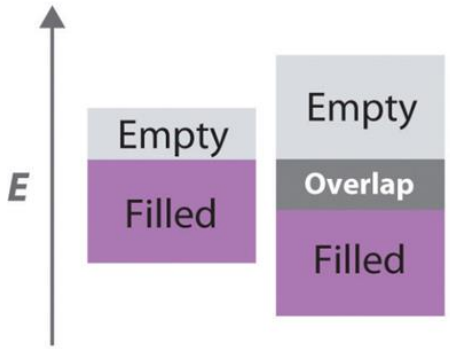
Felix Bloch

Heisenberg's Institute in Leipzig, 1931

Rudolf Peierls Werner Heisenberg

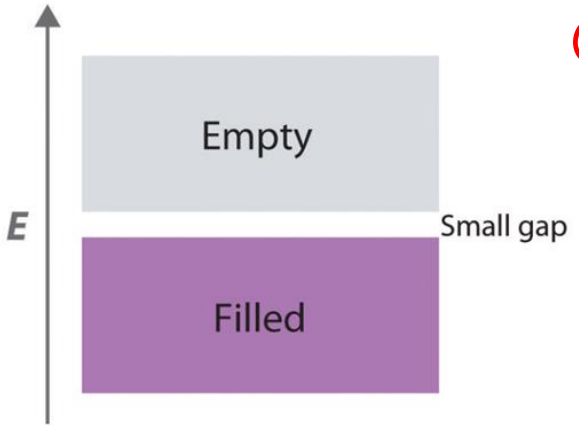
https://repository.aip.org/front-row-l-r-rudolf-peierls-w-heisenberg-back-row-l-r-g-placzek-g-gentile-g-wick-f-bloch-v

Materials classification (Energy band)

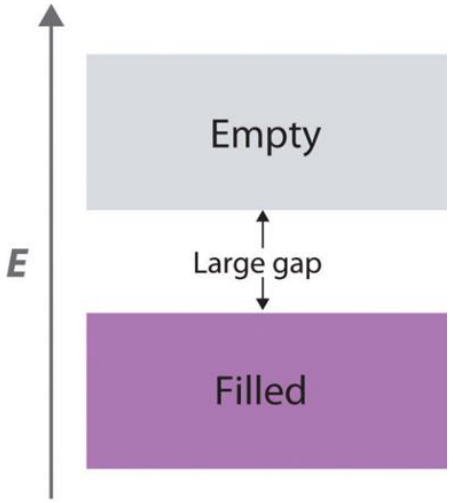


Metals

@ 0K



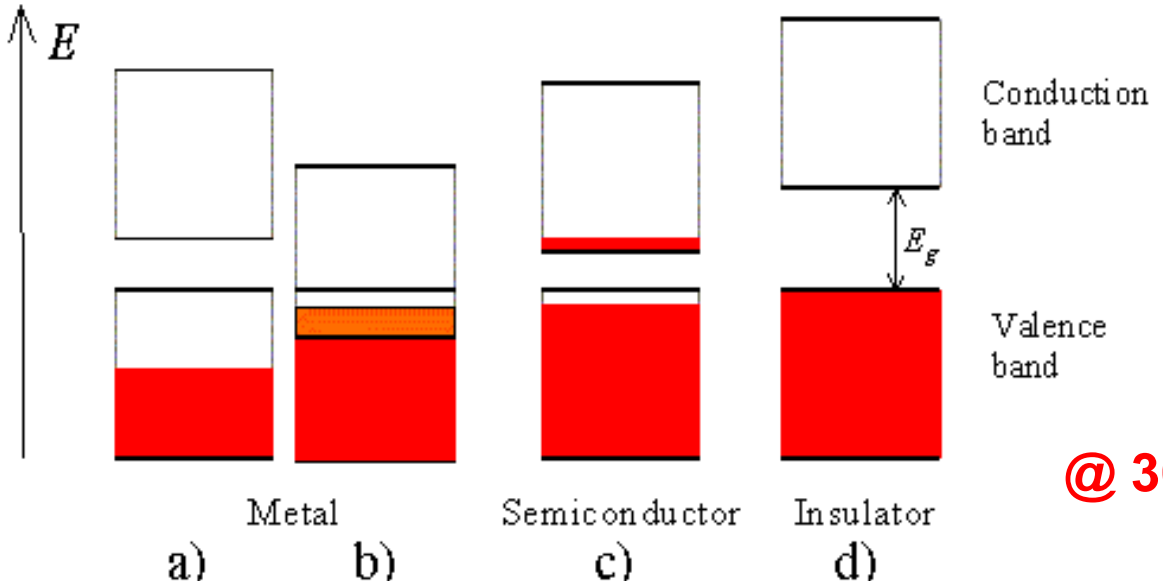
Semiconductors



Insulator s

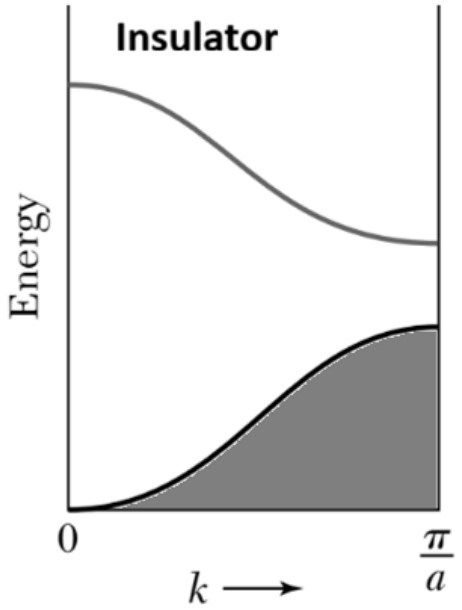
Alan Wilson (1931)

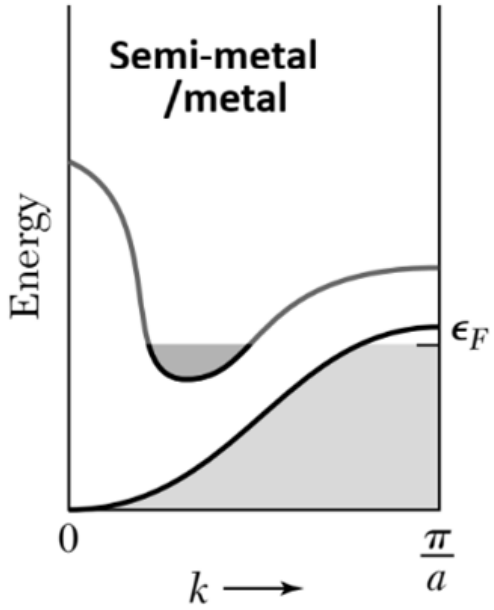


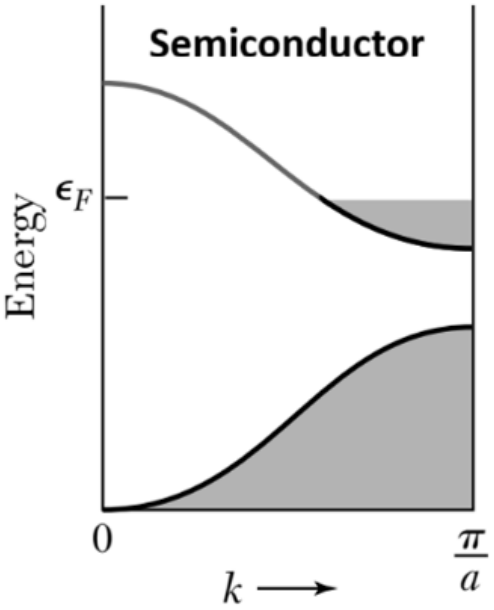


@ 300K

Materials classification (bandstructure)

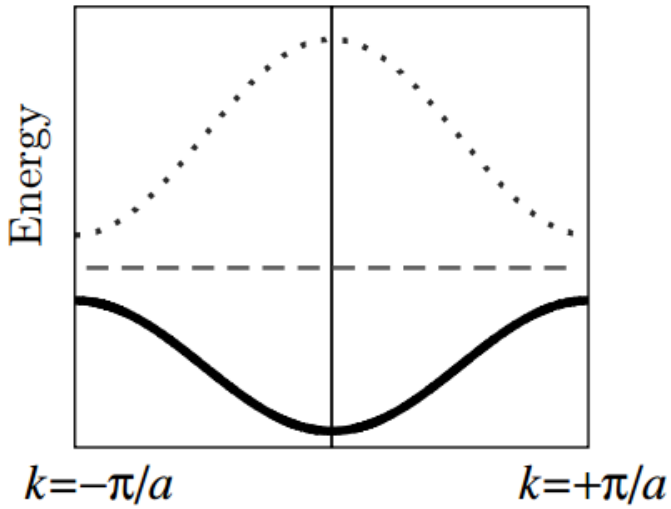




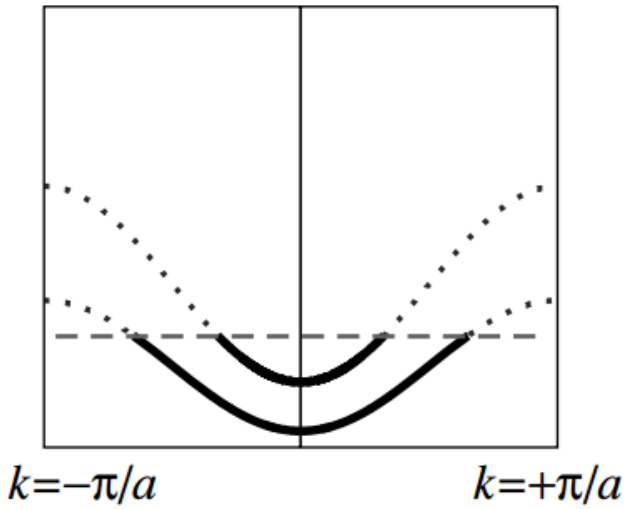


@ 300K

Semiconductor/insulator @ 0K



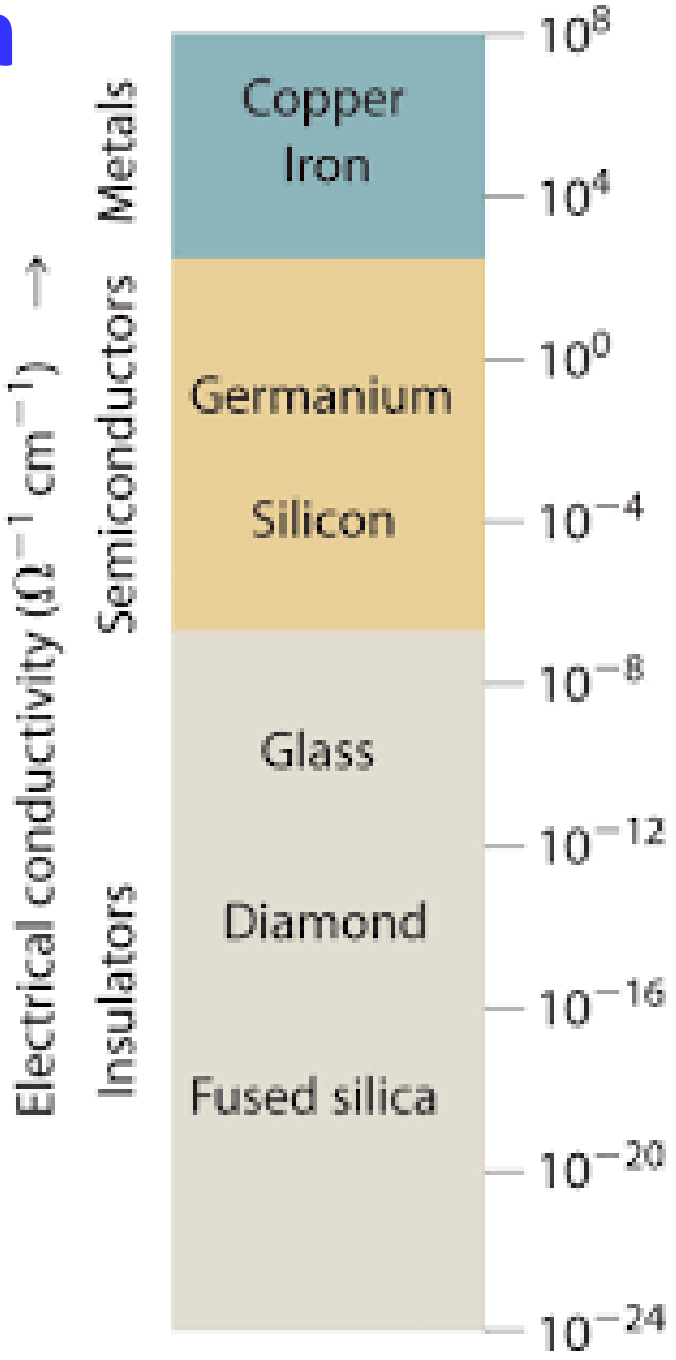
Metal



Clarity on the conductivity conundrum

TABLE 2.1 Energy-Gap Values for Different Materials at 300 K

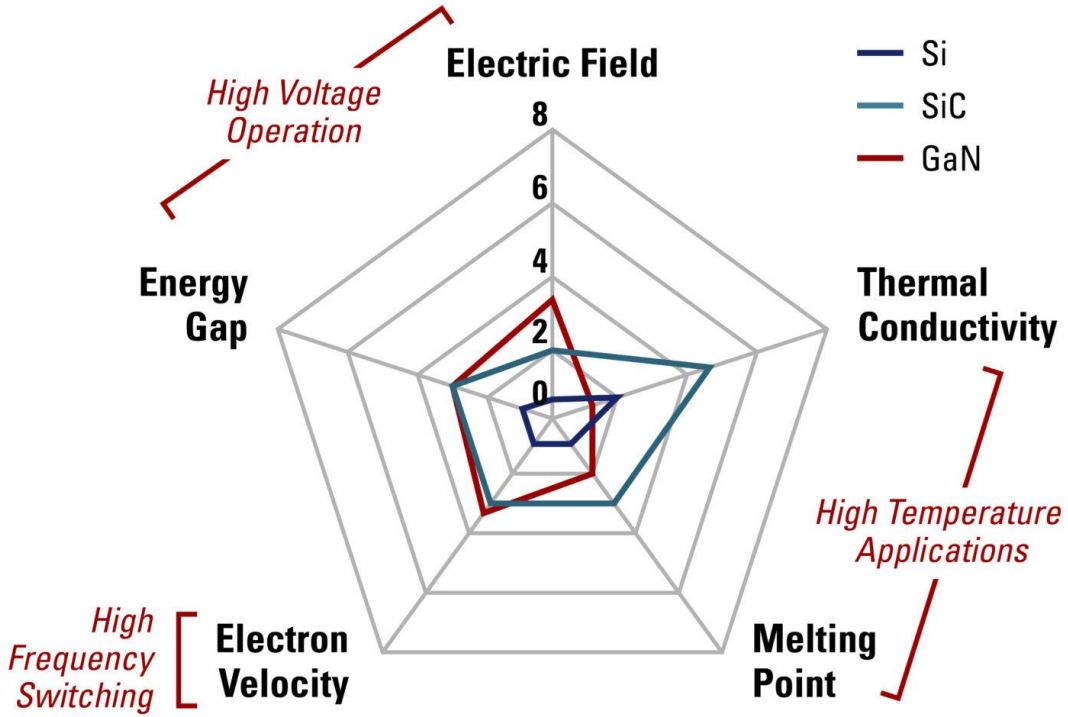
Material	E_g (eV)
Germanium	0.66
Silicon	1.12
Gallium arsenide	1.42
Silicon carbide (cubic)	2.4
Gallium nitride	3.4
Silicon nitride (Si_3N_4)	5
Diamond	5.5
Silicon dioxide (SiO_2)	9

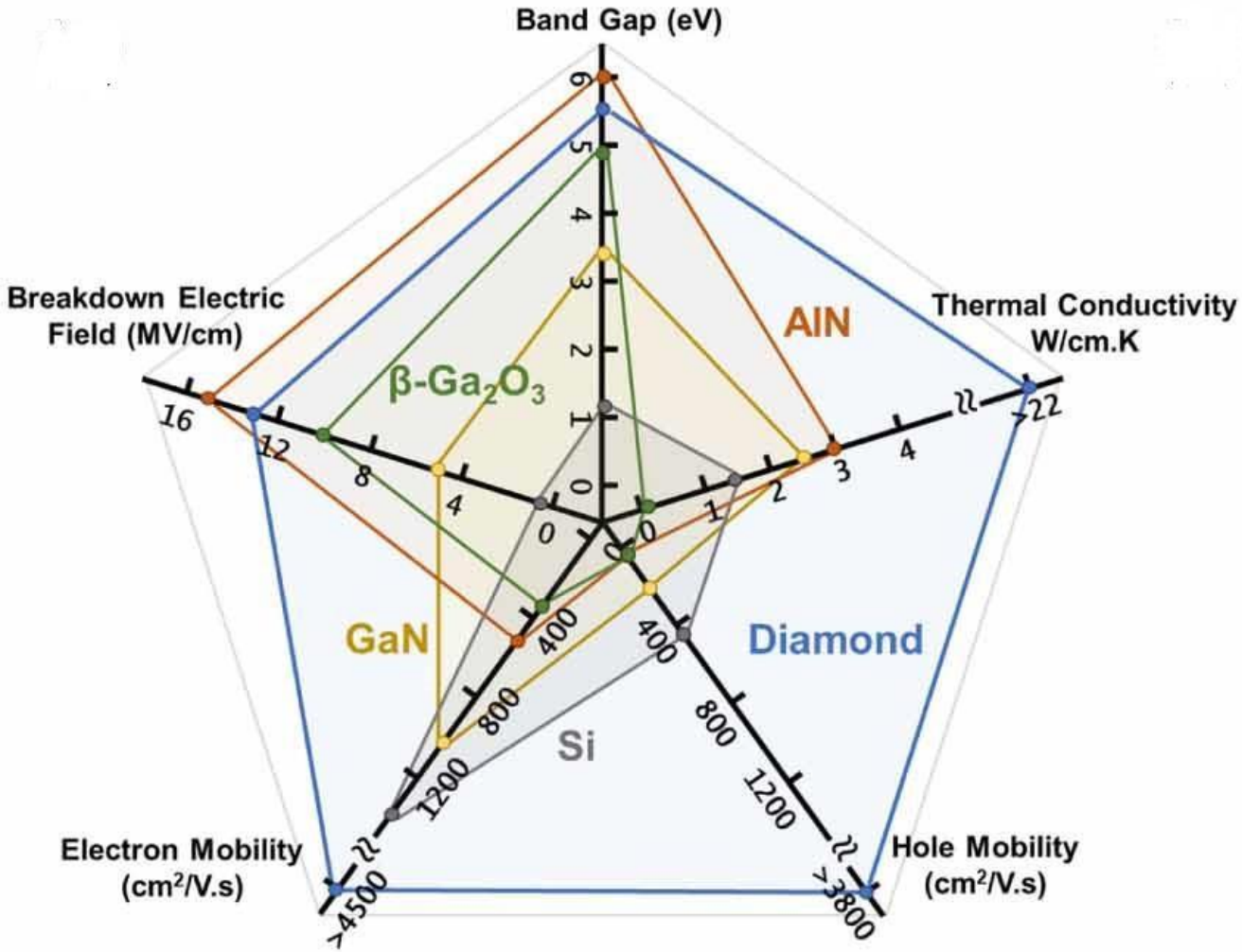


- Why does a material display a specific value for a specific property from the range of values possible for that property?

Wide bandgap and Ultra-wide bandgap semiconductors

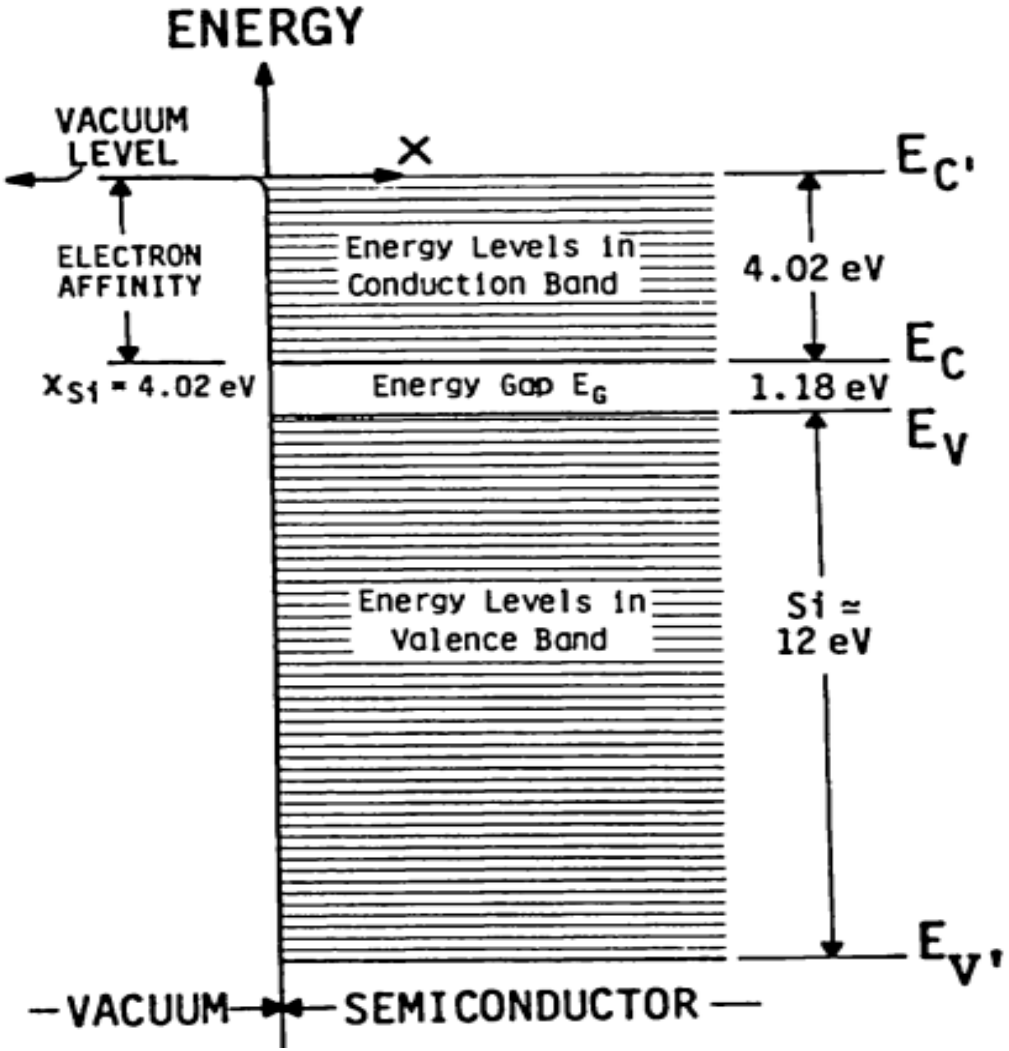
Motivation - light emission in the UV/visible range for optoelectronics and higher breakdown voltages for power electronics

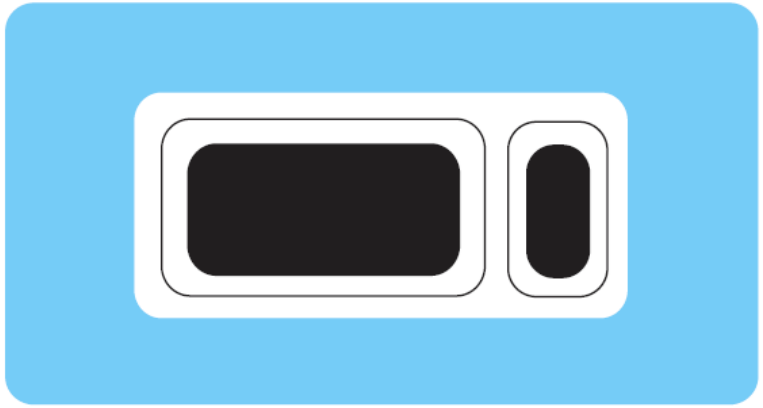




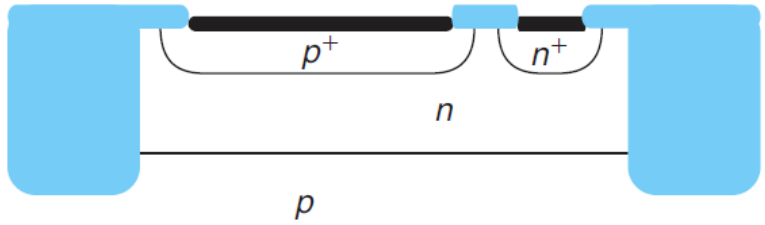
Energy band diagram (E-x)

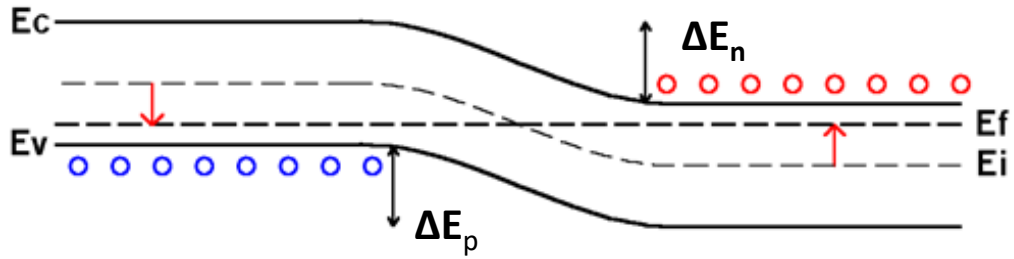
E-x diagram becomes useful when there is an irregularity of material composition and/or electric field in real space (happens in devices)





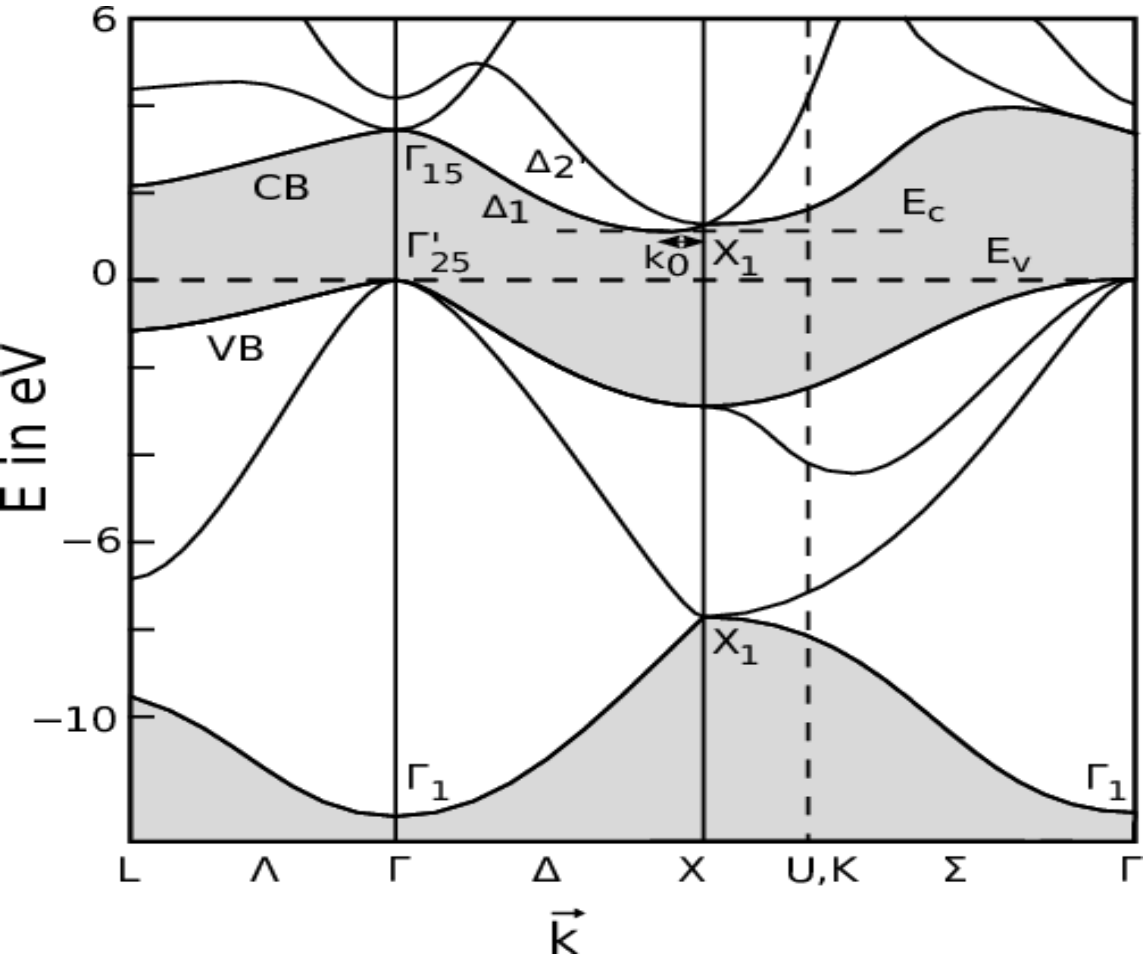
Top view and cross-sectional view of integrated planar PN junction diode



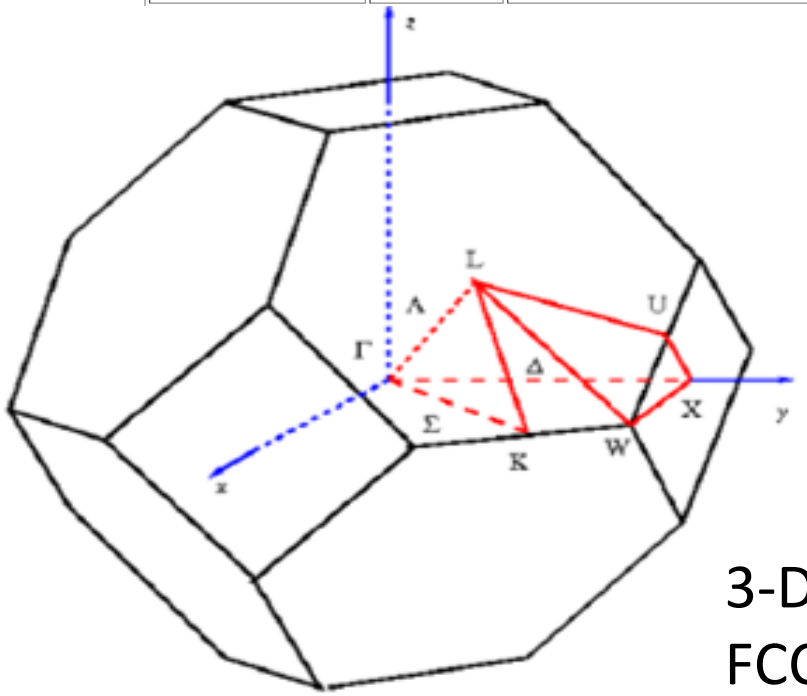


Energy band structure (E-k)

E-k diagram gives a richer and more compact representation in the case of an infinite homogeneous crystal in real space (eg: Optoelectronics)

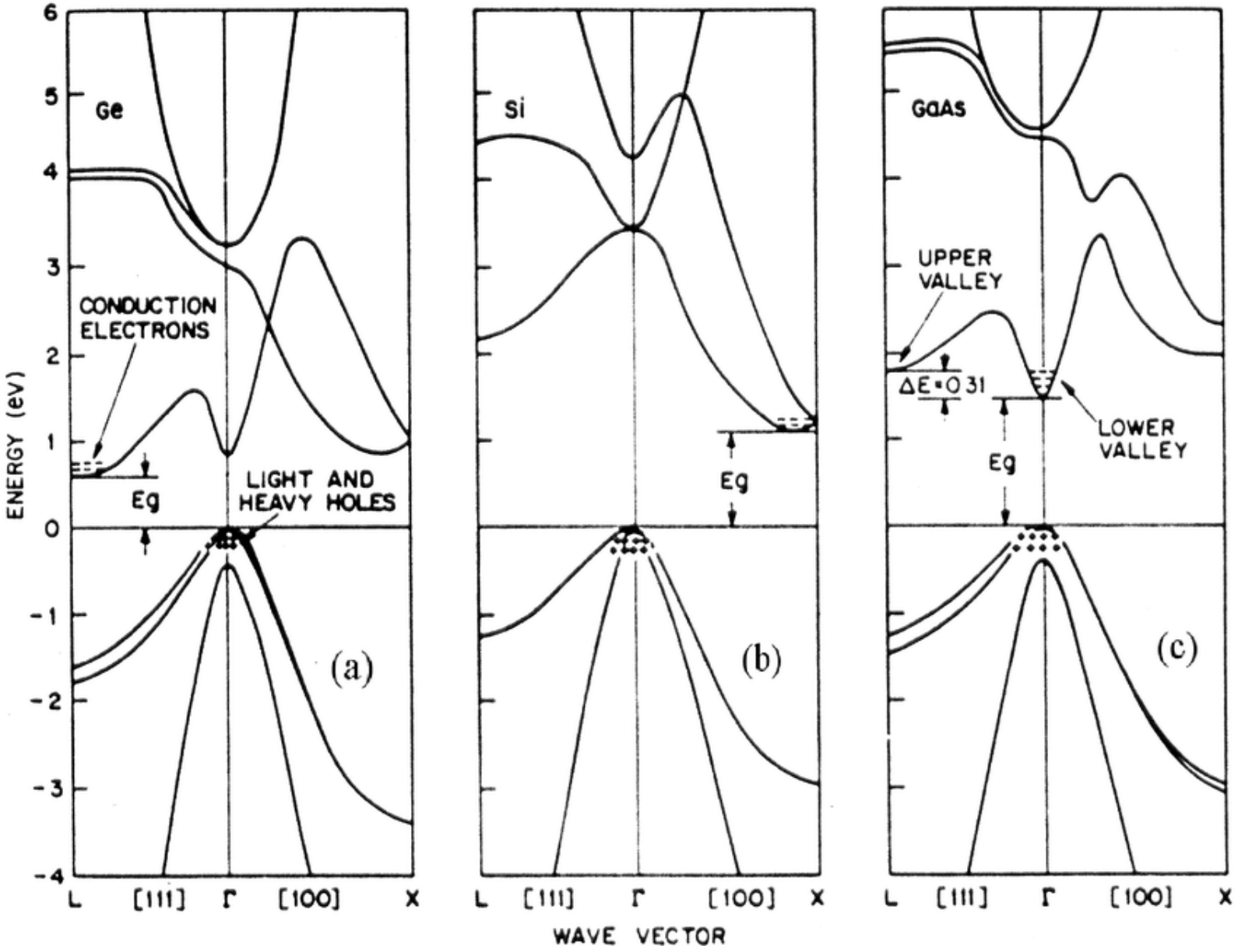


Symmetric Points/ Directions	Coordinate	Remark
Γ	$(0,0,0)$	Origin of **k** space
X	$(1,0,0)$	Middle of square faces
L	$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$	Middle of hexagonal faces
K	$(\frac{3}{4},\frac{3}{4},\frac{3}{4})$	Middle of edge shared by two hexagons
U		Middle of edge shared by a hexagons and a square
W		Middle of edge shared by two hexagons and a square
Δ	$\langle 1,0,0\rangle$	Directed from Γ to X
Λ	$\langle 1,1,1\rangle$	Directed from Γ to L
Σ	$\langle 1,1,0\rangle$	Directed from Γ to K



3-D Brillouin zone of FCC lattice

Direct and indirect semiconductors



CB min and VB max lies
at same k – Direct (GaAs)
at diff k – Indirect (Si, Ge)

Electron-hole recombination and light emission is efficient only in direct bandgap semiconductors

Effective mass of a particle

$$F = m^* \frac{dv_g}{dt} \text{ ------(1)}$$

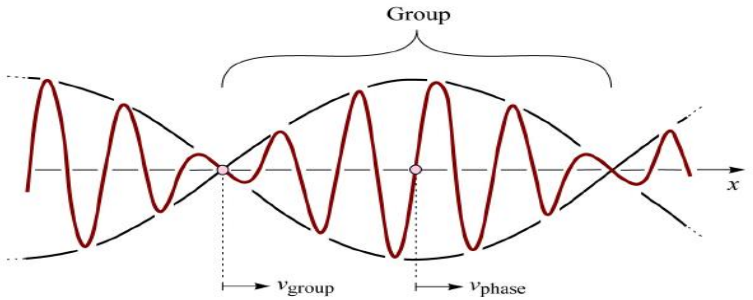
$$F = \hbar \frac{dk}{dt} \text{ -------(2)}$$

Substituting in (1) and equating (1) and (2), we have

$$\hbar \frac{dk}{dt} = m^* \frac{1}{\hbar} \frac{d}{dk} \left(\frac{dE}{dk}\right) \frac{dk}{dt}$$

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \left(\frac{d^2E}{dk^2}\right)$$

Group velocity



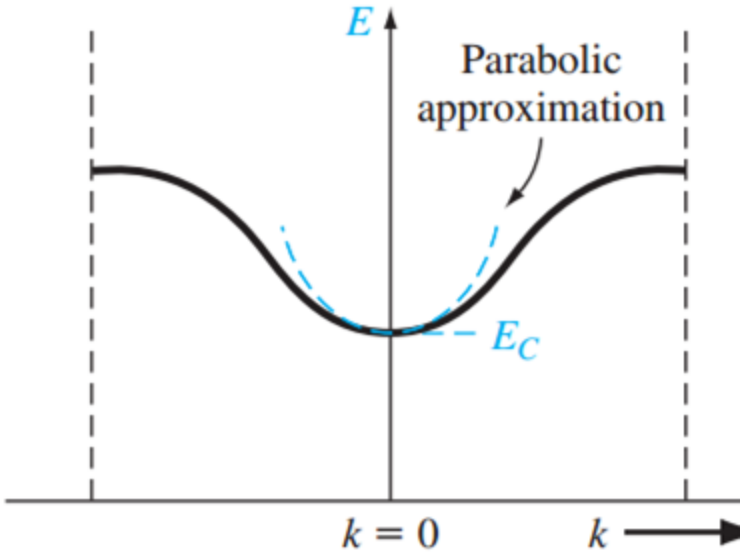
$$v_g = \frac{d\omega}{dk} = \frac{1}{\hbar} \frac{dE}{dk}$$

$$\frac{dv_g}{dt} = \frac{1}{\hbar} \frac{d}{dt} \left(\frac{dE}{dk}\right) = \frac{1}{\hbar} \frac{d}{dk} \left(\frac{dE}{dk}\right) \frac{dk}{dt}$$

Rate of change of crystal momentum ($\hbar$k) = externally applied force

Electrons in a periodic potential behaves as a free electron with a different mass!

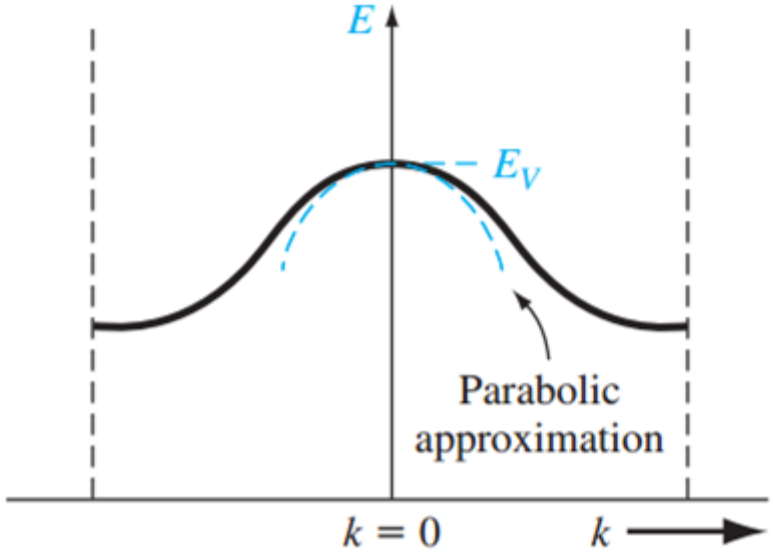
Band structure effective mass of a particle



$$E - E_c = C_1(k)^2$$

$$\frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{2C_1}{\hbar^2} = \frac{1}{m^*}$$

The electron in the bottom of the conduction band can be thought as a classical particle whose motion can be modeled by Newtonian mechanics, provided that effect of the periodic potential and other quantum mechanical properties are taken into account through the effective mass.

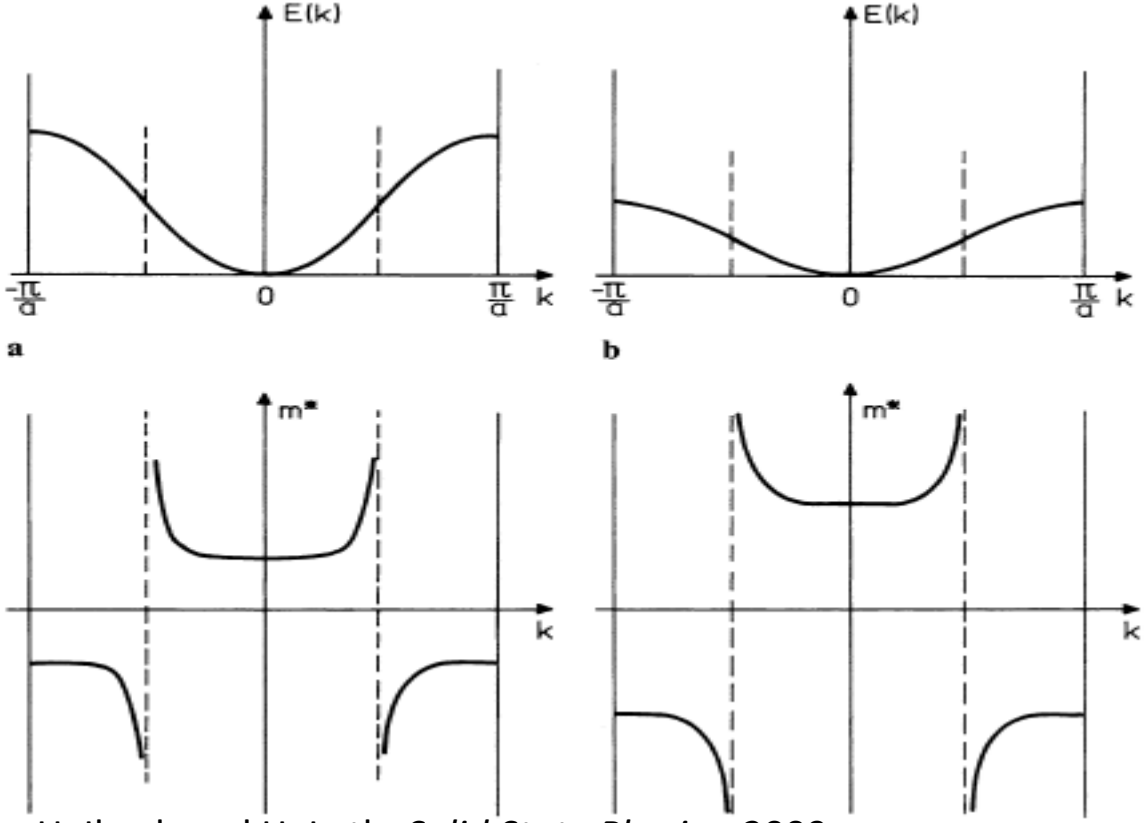


$$(E - E_v) = -C_2(k)^2$$

$$\frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{-2C_2}{\hbar^2} = \frac{1}{m^*}$$

An electron moving near the top of an allowed energy band behaves as if it has a negative mass.

Band structure effective mass of a particle



H. Ibach and H. Luth, *Solid State Physics*, 2009

$$\frac{1}{m_{ij}} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_i \partial k_j}$$ **= Curvature of E-k**

m^* is positive near the bottoms of all bands

m^* is negative near the tops of all bands.

- Electron/hole in a crystal can be treated semi-classically as a free particle but with free space mass replaced by effective mass
- Helps to perform device analysis with minimal use of quantum mechanics
- Use appropriate “average” of band structure effective masses to compute density of states (DoS), conductivity/mobility etc

Band structure effective mass of a particle

Band structure effective mass depends on

- **material**
- **energy**
- **direction of propagation.**

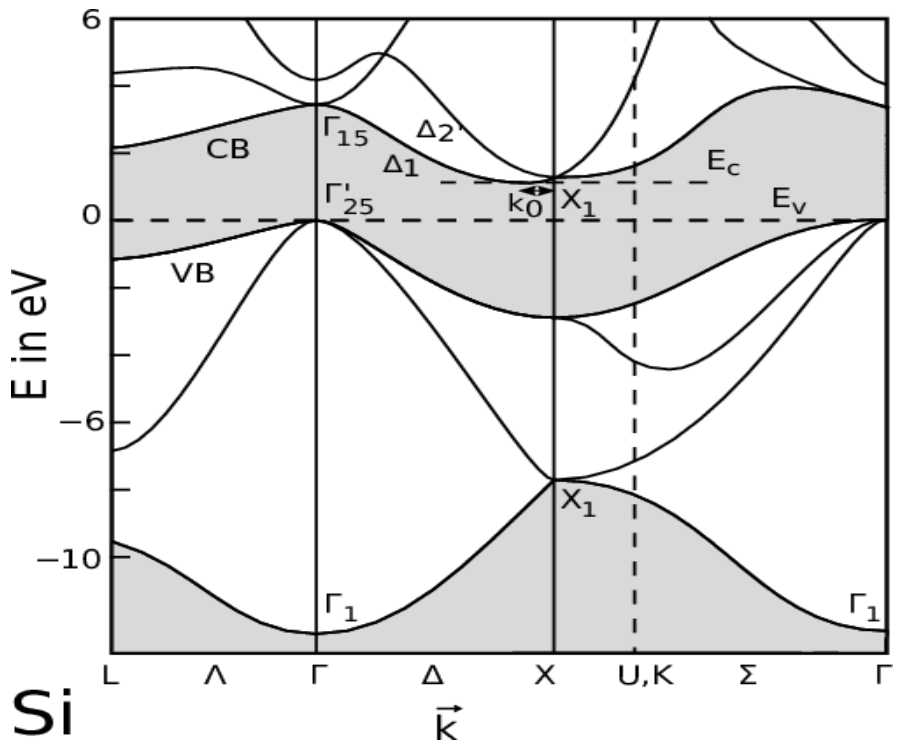
Effective mass is really a tensor

$$\frac{1}{\mathbf{m}^*} = \begin{pmatrix} m_{xx}^{-1} & m_{xy}^{-1} & m_{xz}^{-1} \\ m_{yx}^{-1} & m_{yy}^{-1} & m_{yz}^{-1} \\ m_{zx}^{-1} & m_{zy}^{-1} & m_{zz}^{-1} \end{pmatrix}$$

$$\frac{d\mathbf{v}}{dt} = \frac{1}{\mathbf{m}^*} \cdot \mathbf{F}$$

The crystal does not weigh any less if the effective mass of some of the electrons carrier is less or negative than the free electron mass

It is a relationship between how an electron in a periodic potential is accelerated relative to the lattice in an applied electric or magnetic field



Comparison of a missing electron in VB & a hole

Missing electron (near top of VB)	Assignment for Hole
-ve charge	+ve charge
-ve energy (top of VB is taken as 0 eV)	+ve energy (energy of hole increases downwards)
Wavenumber, k_e	$k_h = -k_e$
Group velocity, v_e	$v_h = v_e$
-ve effective mass	+ve effective mass
Valence band electrons are "real". They can be emitted to vacuum*	Holes are Quasi particles. Holes cannot be emitted to vacuum. Their effect can only be observed while they are inside VB of a solid. Holes can be moved from the VB of one material/region to VB of another material/region. (This is how current is flown in PMOS transistors)

*M. Ding, H. Kim, A.I. Akinwande, "**Observation of valence band electron emission from n-type silicon field emitter arrays**" *Appl. Phys. Lett., 75 (1999), pp. 823-825*

Comparison of a missing electron in VB & a hole

1. $$\mathbf{k}_h = -\mathbf{k}_e \ . \qquad (17)$$

The total wavevector of the electrons in a filled band is zero: $\Sigma\mathbf{k} = 0$, where the sum is over all states in a Brillouin zone. This result follows from the geometrical symmetry of the Brillouin zone: every fundamental lattice type has symmetry under the inversion operation $\mathbf{r} \rightarrow -\mathbf{r}$ about any lattice point; it follows that the Brillouin zone of the lattice also has inversion symmetry. If the band is filled all pairs of orbitals $\mathbf{k}$ and $-\mathbf{k}$ are filled, and the total wavevector is zero.

If an electron is missing from an orbital of wavevector $\mathbf{k}_e$, the total wavevector of the system is $-\mathbf{k}_e$ and is attributed to the hole. This result is surprising: the electron is missing from $\mathbf{k}_e$ and the position of the hole is usually indicated graphically as situated at $\mathbf{k}_e$, as in Fig. 7. But the true wavevector $\mathbf{k}_h$ of the hole is $-\mathbf{k}_e$, which is the wavevector of the point G if the hole is at E. The wavevector $-\mathbf{k}_e$ enters into selection rules for photon absorption.

The hole is an alternate description of a band with one missing electron, and we either say that the hole has wavevector $-\mathbf{k}_e$ or that the band with one missing electron has total wavevector $-\mathbf{k}_e$.

2. $$\epsilon_h(\mathbf{k}_h) = -\epsilon_e(\mathbf{k}_e) \ . \qquad (18)$$

Here the zero of energy of the valence band is at the top of the band. The lower in the band the missing electron lies, the higher the energy of the system. The energy of the hole is opposite in sign to the energy of the missing electron, because it takes more work to remove an electron from a low orbital than from a high orbital. Thus if the band is symmetric,[1] $\epsilon_e(\mathbf{k}_e) = \epsilon_e(-\mathbf{k}_e) = -\epsilon_h(-\mathbf{k}_e) = -\epsilon_h(\mathbf{k}_h)$. We construct in Fig. 8 a band scheme to represent the properties of a hole. This hole band is a helpful representation because it appears right side up.

3. $$\mathbf{v}_h = \mathbf{v}_e \ . \qquad (19)$$

The velocity of the hole is equal to the velocity of the missing electron. From Fig. 8 we see that $\nabla\epsilon_h(\boldsymbol{k}_h) = \nabla\epsilon_e(\boldsymbol{k}_e)$, so that $\boldsymbol{v}_h(\boldsymbol{k}_h) = \boldsymbol{v}_e(\boldsymbol{k}_e)$.

4. $$m_h = -m_e \ . \tag{20}$$

We show below that the effective mass is inversely proportional to the curvature $d^2\epsilon/dk^2$, and for the hole band this has the opposite sign to that for an electron in the valence band. Near the top of the valence band m_e is negative, so that m_h is positive.

5. $$\hbar \frac{d\mathbf{k}_h}{dt} = e(\mathbf{E} + \mathbf{v}_h \times \mathbf{B}) \ . \tag{21}$$

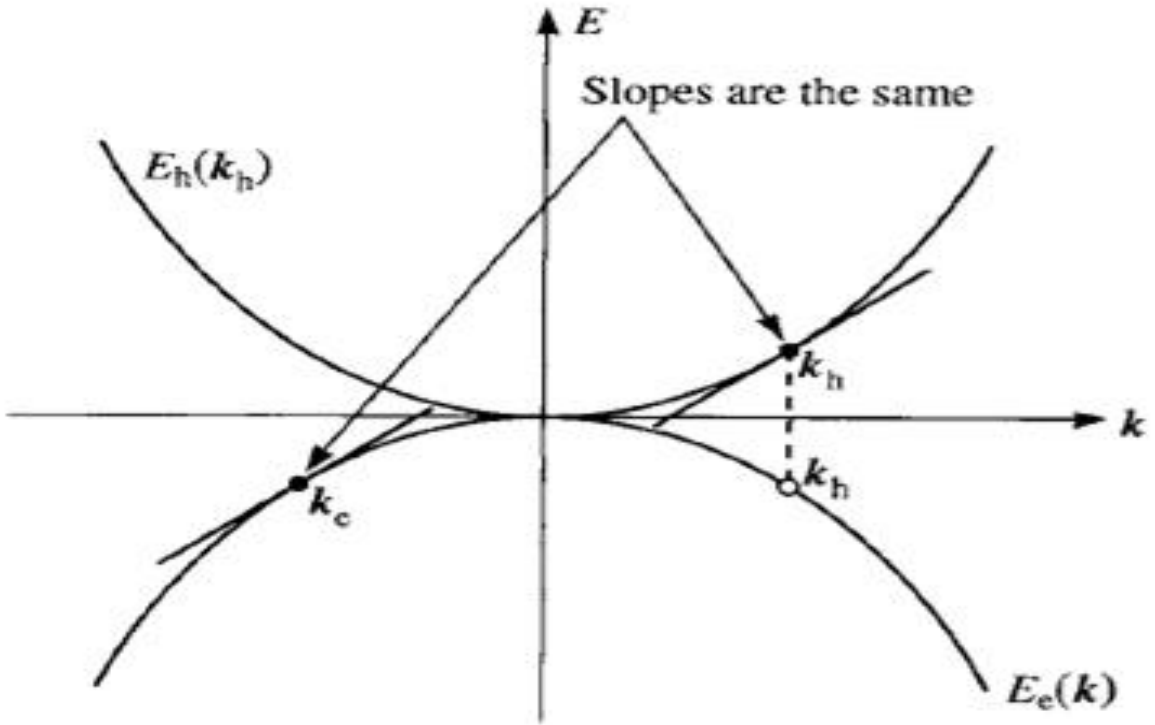
This comes from the equation of motion

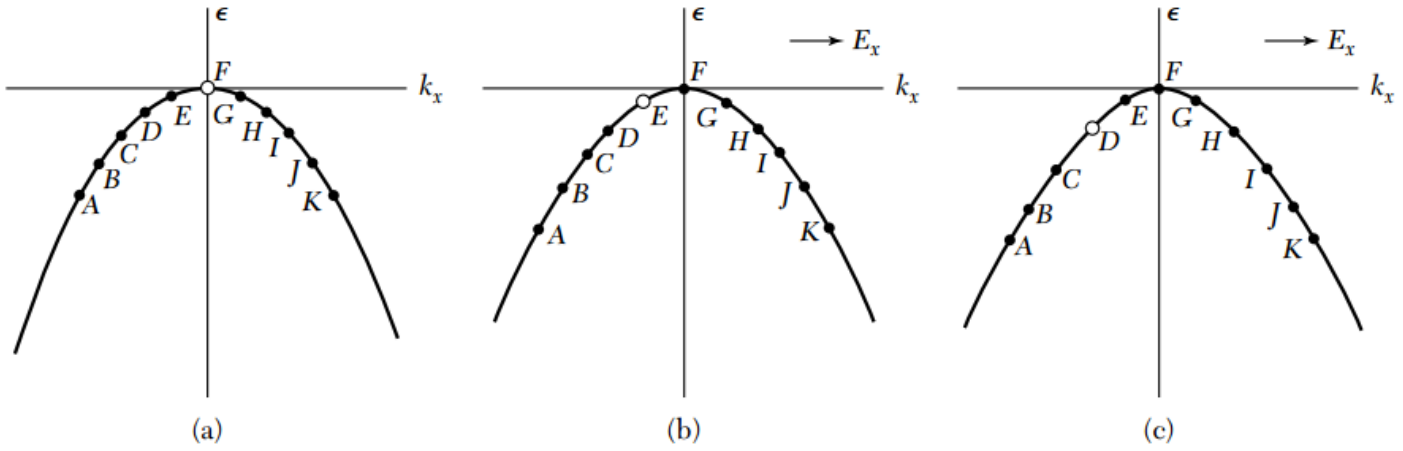
$$\hbar \frac{d\mathbf{k}_e}{dt} = -e(\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) \tag{22}$$

that applies to the missing electron when we substitute $-\mathbf{k}_h$ for $\mathbf{k}_e$ and $\mathbf{v}_h$ for $\mathbf{v}_e$. **The equation of motion for a hole is that of a particle of positive charge *e*.** The positive charge is consistent with the electric current carried by the valence band of Fig. 9: the current is carried by the unpaired electron in the orbital *G*:

$$\mathbf{j} = (-e)\mathbf{v}(G) = (-e)[-\mathbf{v}(E)] = e\mathbf{v}(E) \ , \tag{23}$$

which is just the current of a positive charge moving with the velocity ascribed to the missing electron at *E*. The current is shown in Fig. 10.





For electrons, $\hbar \frac{dk}{dt} = -eE$

$$k(t) = k_0 - \frac{eEt}{\hbar}$$

$$v(k(t)) = v\left(k_0 - \frac{eEt}{\hbar}\right)$$

$$x = x_0 + \int_0^t v\left(k_0 - \frac{eEt}{\hbar}\right)dt$$

At t =0, all states are filled except F at the top of the band; the velocity v_x is zero at F because $d\varepsilon/dk_x = 0$. (b) The force on the electrons is in the $-k_x$ direction and all electrons make transitions together in the $-k_x$ direction, moving the hole to the state E. (c) After a further interval (with no scattering) the electrons move farther along in k space and the hole is now at D. If there is no scattering the hole continues to move down the E-k diagram **(Corrected the figure caption in Charles Kittel)**

where $v(k) = \frac{1}{\hbar}\frac{dE(k)}{dk}$

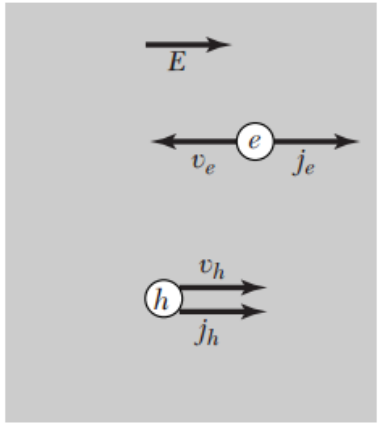


Figure 10 Motion of electrons in the conduction band and holes in the valence band in the electric field *E*. The hole and electron drift velocities are in opposite directions, but their electric currents are in the same direction, the direction of the electric field.